Happiest Health Digital Twin Challenge 2026

NoctTwin

A personalised digital twin for nocturnal hypoglycemia in type 2 diabetes

Team AlooDUM · Arka Jain University · Abhinav Tiwary · Nivesh Kumar · Kriti Kumari · Bhoomi Singh

The problem

Lows that clinic visits miss

A sleeping patient cannot react to falling glucose. A clinic visit sees fasting glucose and HbA1c, not the night in between.

Silent at night

Nocturnal hypoglycemia often goes unnoticed by the patient.

Too many readings

CGM records the night, but thousands of readings are hard to review.

A short list

A clinician needs to know who is at risk tonight, and why.

Use case and data

Bedtime risk of a persistent low

At 22:00, estimate the risk of two or more consecutive CGM readings below 70 mg/dL between 23:00 and 06:00.

100
hospitalised patients

1,181
patient-nights

124
persistent lows (10.5%)

ShanghaiT2DM (Zhao et al., Scientific Data, 2023, CC BY 4.0). Dynamic stream: CGM every 15 minutes. Static stream: labs, medications, demographics. Anonymised public data only, in line with the DPDP Act and HIPAA.

How the twin works

One night, in three moments

18:00

Learn

Earlier nights, CGM, medications and labs build a personal overnight profile. Nothing after 22:00 is used.

22:00

Synchronise

The forecast starts from the actual reading, then relaxes toward the patient's own profile.

23:00 to 06:00

Estimate and explore

Calibrated risk, an uncertainty range, and what-if scenarios with assumed effects.

Calibrated logistic regression, validated by patient. Python, scikit-learn, SHAP, FastAPI, React and TypeScript.

Result 1

Personal history improves prediction

0.74 → 0.83

AUROC, CGM alone to CGM plus the patient's earlier nights.
Ranges over 20 repeated runs: 0.71 to 0.75, and 0.82 to 0.84.
Paired gain +0.10, 95% CI +0.05 to +0.15.

Metric	CGM only	With history
AUROC	0.74	0.83
AUPRC	0.35	0.48
Brier score	0.073	0.065

Nights with earlier history: 1,081 nights, 101 events. Patient-grouped 5-fold validation, 20 seeds. Exploratory, with no external validation.

Result 2

Forecast error and uncertainty

Horizon	Synchronised	Hold bedtime value
1 hour	14.8	18.5
3 hours	19.4	32.0
6 hours	16.1	37.0

Mean absolute error, mg/dL. The patient's own average profile is close at 6 hours (17.2).

±26 mg/dL

80% range. Held-out coverage 79.9%.

±38 mg/dL

90% range. Held-out coverage 89.8%, lower near the 3-hour mark at 86.3%.

Honest findings

What did not help, reported as found

Static data

Labs, demographics and medication tags added no measurable gain over CGM and history.

First-night anomaly

23.0% of first nights had a low, against 9.3% later. The cause is unresolved.

Concentrated events

10% of patients hold 48% of the persistent lows.

Sulfonylurea subgroup

Exploratory: 18 patients, 11 with events. No causal claim.

Clinical meaning

Alert burden

1 in 3

alerts is a true persistent low at 80% specificity, with about 1.8 alerts per patient-week.

A precision of 75% is out of reach at safe sensitivity when the event rate is 10.5%.

NoctTwin is a screening and prioritisation aid, not a diagnostic device.

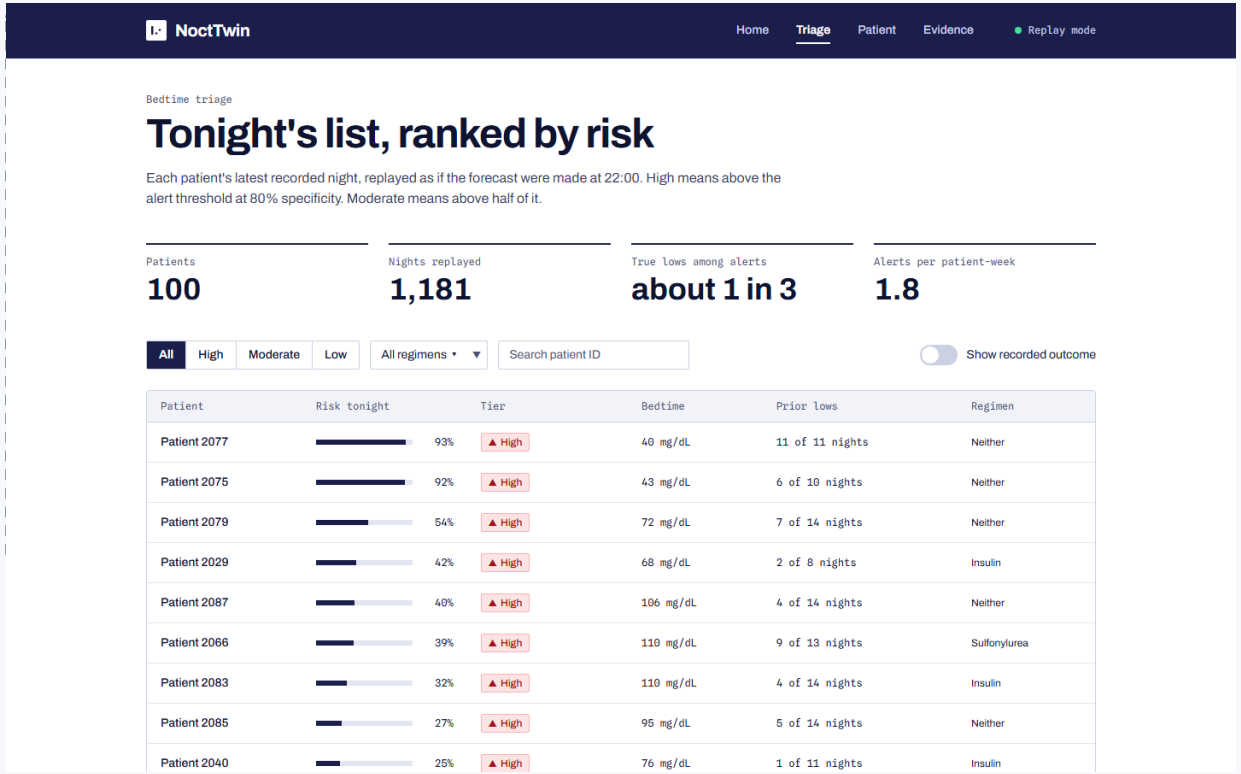
The clinician's view

Tonight's list, and the reasons

Triage board. Patients ranked by calibrated risk, with tiers tied to the alert threshold.

Patient twin. Recorded history, expected profile, uncertainty range, factors behind the flag, and what-if scenarios.

Replay mode. Recorded outcomes stay hidden by default.



Limits and outlook

What we state up front, and what comes next

Limits

Hospitalised cohort and a small sample. No external validation. Regimen labels are per patient. What-if effects are assumptions.

Next steps

Validate on outpatient and Indian cohorts. Test prospectively. Gather clinician feedback on alert burden.

Code: github.com/abhinavtiwary15/AlooDUM_ArkaJainUniversity · MIT licence · Data: ShanghaiT2DM, CC BY 4.0